# Information-Theoretic Analysis of Rule Discovery in ASF Tape Validation Systems

Generated Analysis for tape-whisperer

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**Abstract**

This document presents a comprehensive information-theoretic framework for analyzing the rule discovery problem in Asset-Backed Securities (ASF) tape validation systems. We derive entropy bounds for individual cells, analyze the theoretical maximum number of validation rules, and establish practical quality thresholds based on Minimum Description Length (MDL) principles. Our analysis shows that while the theoretical rule space exceeds $10^{12}$ possibilities, practical quality filtering reduces this to approximately 3,000–5,000 useful rules, representing a compression ratio of $10^8$:1.

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# 1 Introduction

Automated validation of financial spreadsheet data requires discovering patterns and rules that govern the relationships between fields. In the context of ASF (American Securitization Forum) tapes—standardized spreadsheets containing loan-level data—these rules capture domain knowledge about valid data combinations, conditional dependencies, and business logic constraints.

This analysis addresses three fundamental questions:

1. What is the entropy of a single cell in an ASF tape?
2. How many potential validation rules exist for a spreadsheet with 224 columns?
3. What is the practical upper bound on useful rules after quality filtering?

# 2 Single Cell Entropy Analysis

## 2.1 Shannon Entropy Fundamentals

For a discrete random variable $X$ with possible values $\{x_1, x_2, \ldots, x_n\}$ and probability distribution $p(x_i)$, the Shannon entropy is defined as:

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log_2 p(x_i) \tag{1}$$

This measures the average information content (in bits) of the variable.

## 2.2 Cell Type Entropy

Consider a cell that can contain one of three data types: alphabet (string), positive float, or datetime. Assuming equal probability:

$$H_{\text{type}} = -3 \times \frac{1}{3} \log_2 \left(\frac{1}{3}\right) = \log_2(3) \approx 1.585 \text{ bits} \tag{2}$$

Table 1: Entropy for different probability distributions over 3 types

| **Distribution** | **Probabilities** | **Entropy (bits)** |
|---|---|---|
| Uniform | $(0.33, 0.33, 0.33)$ | 1.585 |
| Skewed | $(0.60, 0.30, 0.10)$ | 1.295 |
| Highly skewed | $(0.80, 0.15, 0.05)$ | 0.884 |
| Deterministic | $(1.00, 0.00, 0.00)$ | 0.000 |

## 2.3 Value Entropy by Field Type

Beyond type entropy, each cell carries *value entropy* depending on the field:

The total entropy per cell ranges from 1 bit (boolean fields) to over 20 bits (high-cardinality fields), with an average of approximately 10–15 bits across typical ASF columns.

Table 2: Value entropy for typical ASF fields

| **Field Type** | **Cardinality** | **Entropy (bits)** |
|---|---|---|
| Deal ID (unique) | 50,000 | $\approx 15.6$ |
| State code | 50 | $\approx 5.6$ |
| Loan amount | 1,000,000 | $\approx 20.0$ |
| Rate type | 5 | $\approx 2.3$ |
| Boolean flag | 2 | 1.0 |
| Date field (30yr span) | 10,950 | $\approx 13.4$ |

# 3 Spreadsheet-Level Entropy

## 3.1 Total Independent Entropy

For a spreadsheet with $R$ rows and $C$ columns, assuming cell independence:

$$H_{\text{total}} = R \times C \times \bar{H}_{\text{cell}} \tag{3}$$

For an ASF tape with 100 rows and 224 columns:

$$\text{Cells} = 100 \times 224 = 22{,}400 \tag{4}$$

$$H_{\text{type-only}} = 22{,}400 \times \log_2(3) \approx 35{,}502 \text{ bits} \tag{5}$$

## 3.2 Structured Data Has Lower Entropy

In practice, ASF spreadsheets exhibit strong structure:

- Column A is always dates
- Column B is always identifiers
- Columns follow a fixed schema

This structure introduces dependencies that drastically reduce actual entropy. The *conditional entropy* captures what remains:

$$H(\text{Cell}_j|\text{Schema}) \ll H(\text{Cell}_j) \tag{6}$$

# 4 Rule Space Analysis

## 4.1 Association Rule Framework

An association rule takes the form:

$$\text{IF } A_1 = v_1 \wedge A_2 = v_2 \wedge \cdots \wedge A_k = v_k \text{ THEN } B = v_B \tag{7}$$

where $A_1, \ldots, A_k$ are *antecedent* columns and $B$ is the *consequent* column.

## 4.2 Structural Rule Count

The number of possible rule *structures* (ignoring specific values) for $n$ columns with up to $k$ antecedents:

$$N_{\text{structures}}(n, k_{\max}) = \sum_{j=1}^{k_{\max}} n \cdot \binom{n-1}{j} \tag{8}$$

For $n = 224$ columns:

Table 3: Rule structures by antecedent size

| **Max $k$** | **Rule Structures** | $\log_2$ **(bits)** |
|---|---|---|
| 1 | 49,952 | 15.6 |
| 2 | 5,578,496 | 22.4 |
| 3 | 414,315,776 | 28.6 |
| 4 | 23,063,051,456 | 34.4 |
| 5 | $\approx 1.02 \times 10^{12}$ | 39.9 |

### 4.3 Full Rule Space with Values

When including specific value conditions, the rule space explodes:

$$N_{\text{full}}(n, k, c) = \sum_{j=1}^{k} n \cdot \binom{n-1}{j} \cdot c^j \tag{9}$$

where $c$ is the average column cardinality. For $c = 20$:

$$k = 2: \quad \approx 2.2 \times 10^9 \text{ rules} \tag{10}$$

$$k = 3: \quad \approx 1.7 \times 10^{12} \text{ rules} \tag{11}$$

## 5 Information-Theoretic Rule Metrics

### 5.1 Support, Confidence, and Lift

For a rule $A \Rightarrow B$:

$$\text{Support}(A \Rightarrow B) = P(A \cap B) \tag{12}$$

$$\text{Confidence}(A \Rightarrow B) = P(B|A) = \frac{P(A \cap B)}{P(A)} \tag{13}$$

$$\text{Lift}(A \Rightarrow B) = \frac{P(B|A)}{P(B)} = \frac{\text{Confidence}}{P(B)} \tag{14}$$

### 5.2 Pointwise Mutual Information

The information-theoretic equivalent of lift:

$$\text{PMI}(A, B) = \log_2 \frac{P(A, B)}{P(A) \cdot P(B)} \tag{15}$$

A rule with Lift $= 2$ corresponds to PMI $= 1$ bit.

### 5.3 Conditional Entropy Reduction

The core insight: **rules reduce entropy**. A rule's value is:

$$\text{Value}(R) = P(A) \times [H(B) - H(B|A)] = P(A) \times I(A; B) \tag{16}$$

where $I(A; B)$ is the mutual information between antecedent and consequent.

**Theorem 5.1** (Rule Value Bound)**.** The total value of all discovered rules is bounded by the mutual information in the dataset:

$$\sum_{R \in \mathcal{R}} \text{Value}(R) \leq I(\mathbf{X}; \mathbf{Y}) \tag{17}$$

where $\mathbf{X}$ represents antecedent columns and $\mathbf{Y}$ represents consequent columns.

# 6 Minimum Description Length Framework

## 6.1 MDL Principle

The optimal rule set $\mathcal{R}^*$ minimizes:

$$L(\mathcal{R}) + L(D|\mathcal{R}) \tag{18}$$

where:

- $L(\mathcal{R})$ = bits to encode the rules themselves
- $L(D|\mathcal{R})$ = bits to encode data given the rules (exceptions/errors)

## 6.2 Rule Encoding Cost

For a rule with $k$ antecedent columns, average cardinality $c$, and $n$ total columns:

$$L(\text{rule}) = \underbrace{\log_2 \binom{n}{k+1}}_{\text{column selection}} + \underbrace{k \cdot \log_2 c}_{\text{antecedent values}} + \underbrace{\log_2 c}_{\text{consequent value}} \tag{19}$$

For $n = 224$, $c = 20$:

Table 4: Rule encoding cost by antecedent size

| **Antecedents ($k$)** | **Encoding Cost (bits)** |
|---|---|
| 1 | $\approx 23.5$ |
| 2 | $\approx 32.0$ |
| 3 | $\approx 40.5$ |
| 4 | $\approx 49.0$ |

## 6.3 Break-Even Analysis

A rule is worth adding if its encoding cost is less than the entropy it eliminates:

$$L(\text{rule}) < \text{Support} \times N_{\text{rows}} \times \text{Information Gain} \tag{20}$$

# 7 Algorithmic Complexity

## 7.1 Search Space Complexity

where $F$ = frequent itemsets after pruning, $\bar{L}$ = average transaction length.

Table 5: Minimum information gain for rule viability ($k = 2$, cost $\approx 32$ bits)

| **Support** | **Affected Rows** | **Min IG (bits/row)** |
|---|---|---|
| 1% | 1,000 | 0.032 |
| 5% | 5,000 | 0.0064 |
| 10% | 10,000 | 0.0032 |
| 25% | 25,000 | 0.00128 |

Table 6: Computational complexity comparison

| **Algorithm** | **Complexity** | **For ASF (224 cols, 100K rows)** |
|---|---|---|
| Naive exhaustive | $O(R \times 2^C)$ | Infeasible |
| Apriori (5% survival) | $O(R \times F)$ | $\approx 9.4 \times 10^9$ ops |
| FP-Growth | $O(R \times \bar{L} \times k)$ | $\approx 4.7 \times 10^7$ ops |

### 7.2 Time and Memory Estimates

For 100,000 rows and 224 columns:

$$\text{Estimated time (FP-Growth)} \approx 5\text{–}10 \text{ minutes} \tag{21}$$

$$\text{FP-tree memory} \approx 50\text{–}100 \text{ MB} \tag{22}$$

$$\text{Rule storage} \approx 1\text{–}5 \text{ MB} \tag{23}$$

## 8 Practical Rule Bounds

### 8.1 Quality Thresholds

Standard quality thresholds for production rule systems:

- **Minimum support**: 0.1% (100 rows for 100K dataset)
- **Minimum confidence**: 80%
- **Minimum lift**: 1.5×
- **Minimum information gain**: 0.05 bits/row

### 8.2 Filtering Cascade

Table 7: Rule filtering cascade for 224-column ASF tape

| **Stage** | $k = 1$ | $k = 2$ | $k = 3$ |
|---|---|---|---|
| Structural rules | 49,952 | 5.6M | 414M |
| After support filter (5%) | 2,498 | 27,892 | 103,579 |
| After confidence filter (10%) | 250 | 2,789 | 10,358 |
| After info gain filter (50%) | 125 | 1,395 | 5,179 |
| **Final useful rules** | **125** | **1,395** | **5,179** |

### 8.3 Expected Rule Categories

For a tape-whisperer implementation:

Table 8: Expected rule distribution by category

| **Category** | **Count** | **Value** |
|---|---|---|
| Schema/type constraints | $\approx 224$ | HIGH |
| Single-field value dependencies | $\approx 500$ | MEDIUM |
| Pairwise dependencies ($k = 2$) | $\approx 2{,}000$ | MEDIUM |
| Complex interactions ($k = 3$) | $\approx 500$ | HIGH |
| Cross-field validations | $\approx 200$ | HIGH |
| **TOTAL** | $\approx 3{,}500$ | |

## 9 Summary and Conclusions

### 9.1 Key Findings

1. **Per-cell entropy**: 1–20 bits depending on field type, averaging $\approx 12$ bits.
2. **Theoretical rule space**: $\approx 10^{12}$ possible rules for $k \leq 3$ with value conditions.
3. **Information-theoretic bound**: Maximum useful rules limited by dataset entropy ($\approx$ 75M for 1M rows).
4. **Practical rule count**: 3,000–5,000 after quality filtering.
5. **Compression ratio**: $10^8$:1 reduction from theoretical to practical rule space.

### 9.2 Design Implications for tape-whisperer

- **Algorithm choice**: FP-Growth preferred for efficiency ($\approx$ 10 minutes vs. hours for Apriori).
- **Rule storage**: YAML/Git-based storage feasible ($<$ 5 MB for full rule set).
- **Quality metrics**: Use information gain as primary filter; support and confidence as secondary.
- **Incremental updates**: MDL framework enables principled rule addition/removal decisions.

### 9.3 The Core Insight

*Association rule mining is lossy compression of the rule space. Support and confidence are proxies for mutual information and entropy reduction. A rule's value equals its support times its information gain—the expected entropy reduction per row.*

# A Mathematical Derivations

## A.1 Entropy of Uniform Distribution

For $n$ equally likely outcomes:

$$H = -\sum_{i=1}^{n} \frac{1}{n} \log_2 \frac{1}{n} = -n \cdot \frac{1}{n} \log_2 \frac{1}{n} = \log_2 n \tag{24}$$

## A.2 Mutual Information and Rule Value

By definition of mutual information:

$$\begin{aligned} I(A;B) &= H(B) - H(B|A) && (25)\\ &= H(A) - H(A|B) && (26)\\ &= H(A) + H(B) - H(A,B) && (27) \end{aligned}$$

For a rule $A \Rightarrow B$, the expected entropy reduction across all rows is:

$$\mathbb{E}[\Delta H] = P(A) \cdot [H(B) - H(B|A)] = P(A) \cdot I(A;B) \tag{28}$$

This is precisely the rule's "value" in the MDL framework.

# B Python Reference Implementation

```python
import math
from math import log2, comb

def entropy(probs):
    """Shannon entropy in bits."""
    return -sum(p * log2(p) for p in probs if p > 0)

def mutual_information(p_joint, p_a, p_b):
    """Mutual information I(A;B) in bits."""
    if p_joint == 0:
        return 0
    return log2(p_joint / (p_a * p_b))

def rule_encoding_cost(k, n_cols=224, avg_card=20):
    """MDL encoding cost for a rule with k antecedents."""
    col_selection = log2(comb(n_cols, k + 1))
    value_selection = k * log2(avg_card)
    consequent = log2(avg_card)
    return col_selection + value_selection + consequent

def rule_value(support, h_unconditional, h_conditional, n_rows):
    """Total entropy reduction from a rule."""
    info_gain = h_unconditional - h_conditional
    return support * n_rows * info_gain
```

Listing 1: Core entropy calculations